Annex III. Debt Sustainability Analysis[1]

The sharp contraction of the economy and the outsized policy response in the light of the Covid-19 outbreak have led to very large increases of the fiscal deficit and public debt of the UK in 2020. Under the baseline scenario, public debt is projected to remain broadly stable in the medium-term from this year's level as the economy recovers and the emergency measures are unwound. It would land at around 118 percent of GDP (compared to 96 percent of GDP pre-Covid) once economic activity returns to trend. The surge in gross financing needs in 2020 (26 percent of GDP) was covered by a parallel rise in domestic savings. Financing needs return to lower levels thereafter as the fiscal deficit declines. Public debt is sensitive to combined macroeconomic shocks and contingent liabilities simulated in the template. Sustained fiscal restraint for a period of time will be needed to put public debt on a firm downward path.

A. Baseline and Realism of Projections

1. **Baseline fiscal assumptions**. The staff's baseline is built on the medium-term fiscal framework contained in the March 2020 budget, updated with the policies announced after that. In addition, it assumes the authorities implement 0.5 percent of GDP in structural fiscal consolidation in each of FY23 through FY25.

2. **Debt level**. Under the baseline scenario, gross public sector debt is projected to rise from 96 percent in 2019 to over 118 percent of GDP in 2020. The underlying debt continuous rising mildly throughout the forecast period and stabilizes by FY25 reflecting the recovery in economic activity. Public debt remains at 118 percent of GDP after 2025 as the primary balance reaches the debt-stabilizing level (the 5 percent of GDP decline in FY24 and FY25 reflects the unwinding of the BoE's Term Funding Scheme). Net public sector debt stabilizes at around 108 percent of centered GDP).

3. **Fiscal balance and adjustment**. In the baseline projection, the budget deficit worsens in FY20 as a result of the contraction in GDP and the policies in response to the Covid-19 outbreak. A large improvement takes place in FY21 as the economy rebounds and the bulk of the emergency policies dissipate. The deficit continues declining in the following years, as growth remains above potential and consolidation measures are put in place beginning in FY23. The distribution of fiscal adjustment episodes provided in the DSA template (Figure 3) indicates the projected 3-year adjustment in the cyclically-adjusted primary balance of about 11 percent of GDP is unprecedented. However, the figure is distorted by the large additional "cyclical" Covid-related spending and revenue declines, which automatically sunset as the economy recovers. The actual adjustment needed is better approximated by the structural adjustment in FY23–FY25, which at about 2 percent of GDP is well within the range of past experience.

[1] The data are presented on fiscal year basis (April–March) with ratios calculated using fiscal year GDP (not centered-fiscal year GDP). Public sector gross debt is defined as net debt plus liquid assets held by general government and non-financial public corporations.

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4. Realism of baseline assumptions. The median forecast errors for key assumptions (real GDP growth, the primary balance and inflation) are mostly within the error band observed for all countries (Figure 3). There has been generally a modest upward bias in the projections.

5. Heatmap and debt profile vulnerabilities. Risks from the debt level are deemed high by DSA standards, as the level of debt exceeds the benchmark of 85 percent of GDP for advanced economies under the baseline and stress scenarios. Gross financing needs of over 26 percent of GDP in FY20 are above the benchmark of 20 percent, but return to below the threshold in the years ahead. However, the risk is mitigated by the fact that domestic creditors (especially banks) have the capacity to absorb the higher financing needs given relatively low levels of exposure to the government (see Annex IV, Table). Debt profile vulnerability indicators are below early warning thresholds. Interest rates and CDS spreads also suggest that markets view debt vulnerabilities as low.

B. Stochastic Simulations

6. The fan charts illustrate the possible evolution of the debt ratio over the medium term, under symmetric and asymmetric distributions of risk based on historical outturns. Under both distributions there is a high level of certainty that debt will remain close to end FY20 levels in the medium term, with risks to the upside. As above, the decline in debt of about 5 percent of GDP in FY24 and FY25 is the result of the unwinding of the BoE's Term Funding Scheme.

C. Stress Tests

7. The DSA suggests that medium-term debt dynamics are moderately sensitive to the macroeconomic shocks simulated by the DSA template.

- **Growth shock**. In this scenario, real output growth rates are lowered by one standard deviation in FY21 and FY22. The cumulative growth shock is 1.2 percent of GDP (reflecting stable growth over the last decade) with a small impact on debt dynamics.

- **Primary balance shock**. This scenario assumes that the government provides additional stimulus to support the economic recovery. In particular, the fiscal deficit is assumed to be 2 and 1 percent of GDP higher than in the baseline in FY21 and FY22, respectively, with gradual consolidation starting in FY23 (as in the baseline). Under these assumptions, the debt-to-GDP ratio rises more rapidly and then stabilizes at around 123 percent of GDP, just above the baseline.

- **Interest rate shock**. In this scenario, a 260-basis point increase in interest rates is assumed from FY21 on. The effective interest rate edges up to 2.1 percentage points by FY25 (about 0.9 percentage point higher than the baseline). Given the long maturity of UK's debt the impact is mild.

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- **Exchange rate shock**. A shock to the exchange rate operates via its pass-through to inflation, as debt is denominated in local currency. A depreciation of 20 percent is assumed for FY21. This reduces the debt ratio as the denominator effect of higher nominal GDP is only partially offset by the debt impact of higher spending on inflation-linked payments. The scenario abstracts from the impact of inflation on other expenditures and revenues (CPI is used to uprate many direct tax thresholds, some benefits and public service pensions). The stock of index-linked gilts represents about 25 percent of debt, making debt sensitive to changes in RPI inflation (OBR Fiscal Risk Report 2017).

- **Combined macro-fiscal scenario**. This scenario aggregates shocks to real growth, the interest rate, and the primary balance. Under these assumptions, the debt-to-GDP ratio remains on an upward trend throughout the forecast period, reaching almost 128 percent of GDP in FY25. Gross financing needs would remain below the 20 percent of GDP threshold.

- **Contingent liability shock**. This scenario assumes that banking sector problems lead to a one-time bail out of the financial sector, raising non-interest expenditure by 3 percent of banking sector assets in FY21. The scenario can also be interpreted to cover the risk related to pandemic-related government guarantees (which amount to about 3 percent of GDP). Real GDP is also reduced by one standard deviation for two years. Under this scenario, the debt-to-GDP ratio would rise to about 138 percent of GDP in FY21 and continue rising modestly thereafter, reaching 133 of GDP in FY25. Gross financing needs would exceed the 20 percent of GDP threshold in FY21.

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Figure 1. United Kingdom: Public Sector Debt Sustainability Analysis (DSA)—Baseline Scenario

(In percent of GDP, unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

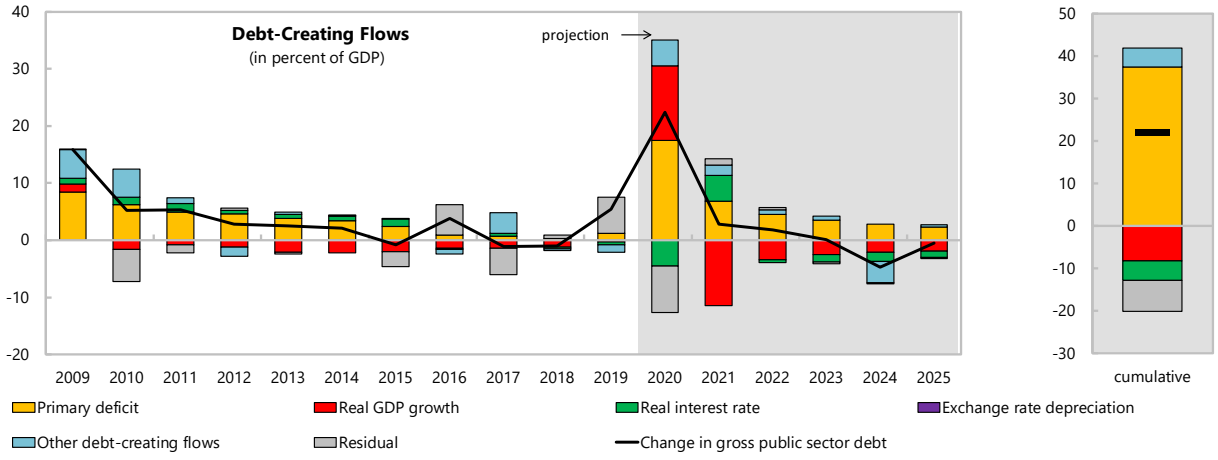
	Actual			Projections					
	2009-2017 [2/]	2018	2019	2020	2021	2022	2023	2024	2025
Nominal gross public debt	85.2	90.6	96.0	118.4	121.2	123.1	123.2	118.4	117.9
Public gross financing needs	11.6	9.4	9.6	26.0	15.6	14.5	13.4	13.7	14.1
Real GDP growth (in percent)	1.5	1.4	0.3	-12.6	10.3	2.9	2.1	1.8	1.6
Inflation (GDP deflator, in percent)	1.7	2.3	2.4	6.4	-3.1	1.1	1.9	2.2	2.2
Nominal GDP growth (in percent)	3.6	3.3	-3.3	0.6	5.6	3.6	4.2	3.9	3.8
Effective interest rate (in percent) [4/]	2.8	2.0	1.9	1.2	0.6	0.7	0.8	0.9	1.2

As of November 19, 2020

Sovereign Spreads		
EMBIG (bp) 3/		63
5Y CDS (bp)		23
Ratings	Foreign	Local
Moody's	Aa3	Aa3
S&Ps	AA	AA
Fitch	AA-	AA-

Contribution to Changes in Public Debt

	Actual			Projections							
	2009-2017	2018	2019	2020	2021	2022	2023	2024	2025	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	4.0	-1.0	5.4	22.4	2.8	1.8	0.1	-4.8	-0.5	21.9	
Identified debt-creating flows	4.9	-1.6	-0.9	30.5	1.7	1.4	0.4	-4.6	-0.3	29.1	
Primary deficit	3.9	0.3	1.3	17.5	6.8	4.5	3.5	2.8	2.4	37.5	-2.7
Primary (noninterest) revenue and grants	36.2	37.0	38.7	35.9	37.2	36.9	37.6	38.1	38.6	224.4	
Primary (noninterest) expenditure	40.2	37.3	40.0	53.4	44.0	41.4	41.1	40.9	41.0	261.9	
Automatic debt dynamics [5/]	-0.4	-1.5	-0.8	8.5	-6.9	-3.9	-3.8	-3.7	-3.0	-12.8	
Interest rate/growth differential [6/]	-0.4	-1.5	-0.8	8.5	-6.9	-3.9	-3.8	-3.7	-3.0	-12.8	
Of which: real interest rate	0.8	-0.3	-0.5	-4.5	4.5	-0.5	-1.3	-1.6	-1.2	-4.6	
Of which: real GDP growth	-1.3	-1.2	-0.3	13.1	-11.4	-3.4	-2.5	-2.1	-1.9	-8.3	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	1.4	-0.3	-1.4	4.5	1.8	0.8	0.8	-3.8	0.3	4.5	
Cash req. adjustments. incl. privatization (negative)	1.4	-0.3	-1.4	4.5	1.8	0.8	0.8	-3.8	0.3	4.5	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	-1.0	0.6	6.3	-8.2	1.1	0.4	-0.3	-0.1	-0.2	-7.3	



Source: IMF staff.

1/ Public sector is defined as consolidated public sector.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

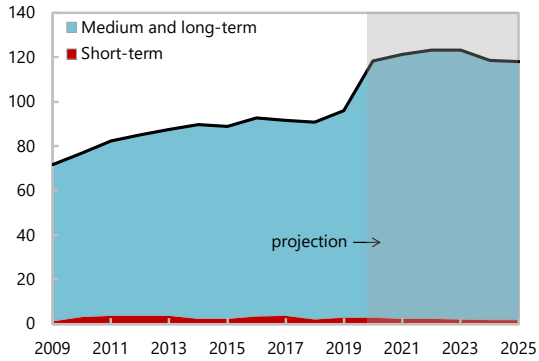
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Figure 2. United Kingdom: Public DSA—Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

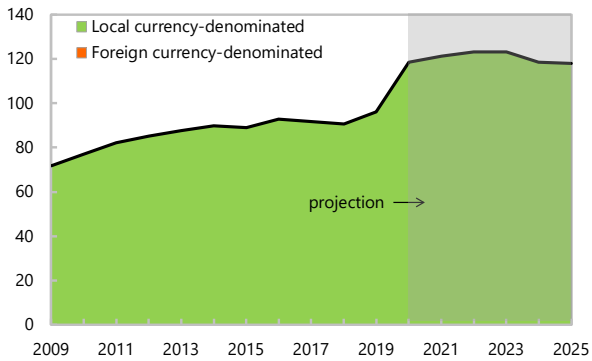
By Maturity

(in percent of GDP)

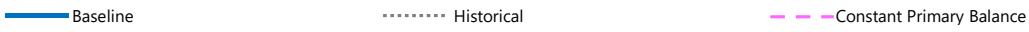


By Currency

(in percent of GDP)

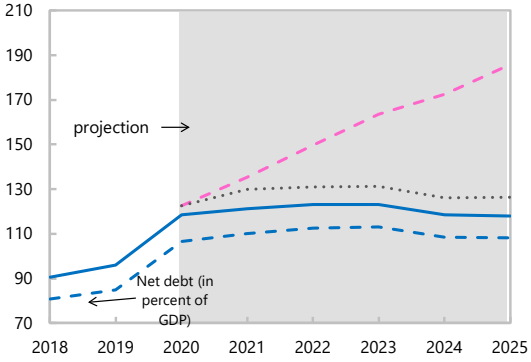


Alternative Scenarios



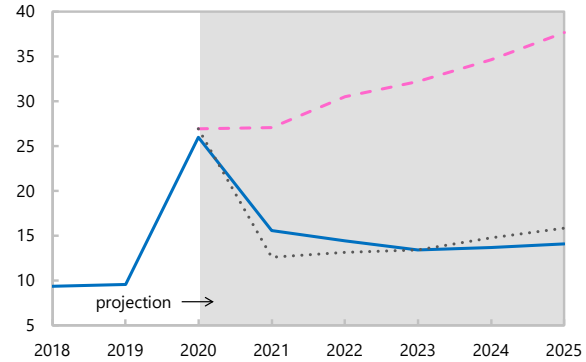
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2020	2021	2022	2023	2024	2025
Real GDP growth	-12.6	10.3	2.9	2.1	1.8	1.6
Inflation	6.4	-3.1	1.1	1.9	2.2	2.2
Primary Balance	-17.5	-6.8	-4.5	-3.5	-2.8	-2.4
Effective interest rate	1.2	0.6	0.7	0.8	0.9	1.2
Constant Primary Balance Scenario						
Real GDP growth	-12.6	10.3	2.9	2.1	1.8	1.6
Inflation	6.4	-3.1	1.1	1.9	2.2	2.2
Primary Balance	-18.1	-18.1	-18.1	-18.1	-18.1	-18.1
Effective interest rate	1.2	0.6	0.7	0.8	0.8	1.0

Historical Scenario	2020	2021	2022	2023	2024	2025
Real GDP growth	-12.6	1.7	1.7	1.7	1.7	1.7
Inflation	6.4	-3.1	1.1	1.9	2.2	2.2
Primary Balance	-18.1	-2.9	-2.9	-2.9	-2.9	-2.9
Effective interest rate	1.2	0.6	0.9	1.1	1.2	1.7

Source: IMF staff.

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